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System Backup

Chapter – 9

Understand the need of Backup

Backup-why

- Risks
  - Humans are quite unreliable; they might make a mistake or be malicious and destroy data on purpose.
  - Modern software does not even pretend to be reliable. A rock-solid program is an exception, not a rule.
  - Nature may not be evil, but, nevertheless, can be very destructive sometimes.
  - The most reliable thing is hardware, but it still breaks seemingly spontaneously, often at the worst possible time.
- Solution
  - To protect yourself against any of these threats, a very cheap and simple control is available: to make regular backups

Backup-What

- Backup should be the result of proper risk analysis.
- Taking full backup of system may not be ; for example, the /proc and /sys filesystems
- The obvious things to back up are user files (/home) and system configuration files (/etc)
- Generally it is good idea to backup everything needed to rebuild the system as fast as required after a failure

Backup-When

- Depending on the rate of change of the data, this may be anything from daily to almost never.
- To determine your minimal backup frequency consider the amount of data you are confident you can afford to loose without severe consequences for you and/or your company.
- Many companies made incremental backups during the night and a full backup during the weekend.
- However, many systems need to be used 24 x 7 which creates the need for alternate strategies, for example creating snapshots on disk, which can subsequently be backed up on tape while the production systems continue their work on the live filesystem.

Backup-How

- Verifying the backup:
  - The safest method is to read back the entire backup and compare this with the original files. This is very time-consuming and often not an option.
  - A faster, and relatively safe method, is to create a table of contents (which should contain a checksum per file) during the backup. Afterwards, read the contents of the tape and compare the two.
- Testing the restore procedure:
  - This means that you must have a restore procedure.
  - This restore procedure has to specify how to restore anything from a single file to the whole system.
  - Every few months, you should test this procedure by doing a restore.

Backup-Where

- If you made that backup on your only medium you lost your data.
- So you should have at least two sets of backup media.
- But if you store both sets in the same building, the first disaster that comes along will destroy all your precious backups along with the running system.
- So you should have at least one set stored at a remote site
- Should have a backup plan documented listing all procedures to follow.
- Example Backup device
  - Tape
  - Disk
  - Optical Media
  - Network Storage

3 Types of Backup

1. Full - A full backup is a complete copy of a business or organization's data assets in their entirety. This process requires all files to be backed up into a single version. It is the best data protection option in terms of speed of recovery and simplicity because it creates a complete copy of the source data set.

2. Differential - A differential backup is a data backup that copies all of the files that have changed since the last full backup was performed. This includes any data that has been created, updated or altered in any way and does not copy all of the data every time

3. Incremental - An incremental backup is a backup type that only copies data that has been changed or created since the previous backup activity was conducted. An incremental backup approach is used when the amount of data that has to be protected is too voluminous to do a full backup of that data every day

Linux Backup Utilities

1. Rsync
2. Tar
3. Dd
4. Cpio

Backup Solutions

1. AMANDA
2. Bacula
3. Bareos
4. BackupPC

Tar (Tape Archive) command - for Backup

- The tar utility is used to combine multiple files and directories into a continuous stream of bytes (and revert the stream intofiles/directories).
- This stream can be compressed, transferred over network connections, saved to a file or streamed onto a tape device.
- When reading files from the stream, permissions, modes, times and other information can be restored.
- Tar is the most basic way for transferring files and directories to and from tape, either with or without compression.
- Options
  - -c create, -v verbose, -p preserve permissions, -z compress, -f specify output file name

Take backup using Tar

```
[root@localhost /]# ls /website/
datastore10.sql  datastore4.sql  tempstore       workfile5.php
datastore11.sql  datastore5.sql  workfile10.php  workfile6.php
datastore12.sql  datastore6.sql  workfile1.php   workfile7.php
datastore1.sql   datastore7.sql  workfile2.php   workfile8.php
datastore2.sql   datastore8.sql  workfile3.php   workfile9.php
datastore3.sql   datastore9.sql  workfile4.php
[root@localhost /]# tar -cvpzf /backup17_10.tar.gz --exclude=/website/tempstore /website
tar: Removing leading `/' from member names
/website/
/website/workfile1.php
/website/workfile2.php
/website/workfile3.php
/website/workfile4.php
/website/workfile5.php
```

Output of tar backup command

```
[root@localhost /]# ll
total 40
-rw-r--r--.   1 root root  372 Oct 17 03:16 backup17 10.tar.gz
lrwxrwxrwx.   1 root root    7 May 10  2019 bin -> usr/bin
dr-xr-xr-x.   6 root root 4096 Aug 16 06:38 boot
drwxr-xr-x.  21 root root 3500 Oct 13 03:13 dev
```

Backup recovery // output directory should be created first

```
[root@localhost /]# tar -xvpzf /backup17_10.tar.gz -C /recover
website/
website/workfile1.php
website/workfile2.php
website/workfile3.php
website/workfile4.php
website/workfile5.php
website/workfile6.php
website/workfile7.php
website/workfile8.php
website/workfile9.php
website/workfile10.php
website/datastore1.sql/
website/datastore2.sql/
```

Use Crontab to backup

- Crontab files are stored in the /var/spool/cron folder.
- The /etc/crontab file and the scripts inside the /etc/cron.d directory are system-wide crontab files that can be edited only by the system administrators.
- You can schedule backup using tar as a cronjob.
- Commands
  - Crontab –e (open up your crontab in your user profile's default text editor)
  - Crontab –l (To view the contents of your crontab)
  - Crontab –r (erase your crontab with the following command)
  - **Command Syntax** **minute hour day month dayofweek command**
- **Example**
  - 15 6 18 * * 0-4 <command to backup>
  - At what time above backup will happen?
  - Complete this activity at home

Backup Using Rsync

- Rsync is a command line backup tool. The Linux rsync command transfers and synchronizes files or directories efficiently between a local machine, another host, remote shell, or any combination of these.
- To install rsync
  - yum install rsync –y
- How to check if rsync is installed
  - rpm –qa rsync
- This will show the rsync package if installed in your system with version number.

Put backup machine and working machine In same network

- Put both machines in internal network, in virtual box settings.
- Go to IPV4 settings in each machines, select "manual" and enter the following IP address.

- Backup Machine IP : 192.168.55.20
- Working Machine IP :192.168.55.10

- Gateway IP : 192.168.55.1
- DNS IP: Set the IP of corresponding system itself.

- Ping should be successful between two systems.

Create some files in /opt directory to backup

- cd /opt

- Then touch to create some files as shown below

```
[root@localhost opt]# touch {A..C}-card-{1..3}.txt
[root@localhost opt]# ls
A-card-1.txt  A-card-3.txt  B-card-2.txt  C-card-1.txt  C-card-3.txt
A-card-2.txt  B-card-1.txt  B-card-3.txt  C-card-2.txt  rh
[root@localhost opt]#
```

Do the following in backup machine

- Go to /opt directory and create a new directory data-backup which will store all backed up data from working system

```
File    Edit    View    Search    Terminal    Help
[root@localhost ~]# cd /opt
[root@localhost opt]# ls
rh
[root@localhost opt]# mkdir data-backup
[root@localhost opt]# ls
data-backup  rh
[root@localhost opt]# cd data-backup/
[root@localhost data-backup]# ls
[root@localhost data-backup]#
```

Run rsync in working machine to push backup to backup server.

```
[root@localhost opt]# rsync -av /opt/ root@192.168.55.20:/opt/data-backup
The authenticity of host '192.168.55.20 (192.168.55.20)' can't be established.
ECDSA key fingerprint is SHA256:GJEqENfMMJPpCAnpINPGothXK77q6SAegyAufteF6xA.
ECDSA key fingerprint is MD5:7f:49:7c:28:f4:d8:b5:77:84:f7:10:c4:a7:22:41:18.
Are you sure you want to continue connecting (yes/no)? y
Please type 'yes' or 'no': yes
Warning: Permanently added '192.168.55.20' (ECDSA) to the list of known hosts.
root@192.168.55.20's password:
sending incremental file list
./
A-card-1.txt
A-card-2.txt
A-card-3.txt
B-card-1.txt
B-card-2.txt
B-card-3.txt
C-card-1.txt
C-card-2.txt
C-card-3.txt
rh/
```

Now open data-backup directory in backup server.

- You can see all files and directories are copied to data-backup folder in backup server

```
[root@localhost ~]# cd /opt
[root@localhost opt]# cd data-backup/
[root@localhost data-backup]# ls
A-card-1.txt  A-card-3.txt  B-card-2.txt  C-card-1.txt  C-card-3.txt
A-card-2.txt  B-card-1.txt  B-card-3.txt  C-card-2.txt  rh
[root@localhost data-backup]#
```

Maintain exact copy of data in backup server use --delete option

create a new file in backup server d.txt in /opt/data-backup which is not available in working server. So that now the backup doesn't exactly match the data in working server.

The following command will erase all differences and make the backup an exact copy of data in working server.

```
[root@localhost opt]# rsync -av --delete /opt/ root@192.168.55.20:/opt/data-backup
root@192.168.55.20's password:
sending incremental file list
./
deleting d.txt

sent 193 bytes  received 32 bytes  50.00 bytes/sec
total size is 0  speedup is 0.00
[root@localhost opt]#
```

You can see the file d.txt is deleted in the backup server

```
[root@localhost data-backup]# ls
A-card-1.txt  A-card-3.txt  B-card-2.txt  C-card-1.txt  C-card-3.txt  rh
A-card-2.txt  B-card-1.txt  B-card-3.txt  C-card-2.txt  d.txt
[root@localhost data-backup]# ls
A-card-1.txt  A-card-3.txt  B-card-2.txt  C-card-1.txt  C-card-3.txt
A-card-2.txt  B-card-1.txt  B-card-3.txt  C-card-2.txt  rh
```

Do the following in working server to get lost data from Backup server

```
[root@localhost opt]# ls
A-card-1.txt  A-card-3.txt  B-card-2.txt  C-card-1.txt  C-card-3.txt
A-card-2.txt  B-card-1.txt  B-card-3.txt  C-card-2.txt  rh
[root@localhost opt]# rm -f A*.txt
[root@localhost opt]# ls
B-card-1.txt  B-card-3.txt  C-card-2.txt  rh
B-card-2.txt  C-card-1.txt  C-card-3.txt
[root@localhost opt]# rsync -av root@192.168.55.20:/opt/data-backup/A*.txt /opt/
root@192.168.55.20's password:
receiving incremental file list
A-card-1.txt
A-card-2.txt
A-card-3.txt

sent 68 bytes  received 181 bytes  55.33 bytes/sec
total size is 0  speedup is 0.00
[root@localhost opt]# ls
A-card-1.txt  A-card-3.txt  B-card-2.txt  C-card-1.txt  C-card-3.txt
A-card-2.txt  B-card-1.txt  B-card-3.txt  C-card-2.txt  rh
[root@localhost opt]#
```

Reference

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